

Table 3 highlights value added modules.

| Name of module         | Description                                                                                                                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qt Automotive Suite    | A collection of software components and tools to enable development of in-vehicle-infotainment (IVI) systems. It is built on top of the 'Qt for Device Creation' offering and is compliant with the GENIVI automotive platform architecture |
| Qt for Automation      | Comprises libraries and tools for automation related domains such as KNX, OPC UA and MQTT                                                                                                                                                   |
| Qt for Device Creation | Offers tools for fast, easy and fully integrated embedded device application development                                                                                                                                                    |

in some Qt frameworks. Unlike the debugger, GammaRay understands those internal structures and can present them in a meaningful way, making it a tool that should not be missing in any Qt developer's toolbox.

**Latest version:** 2.12.0

**Official website:** <https://www.kdab.com/development-resources/qt-tools/gammaray/>

## 2. Clazy

Clazy helps Clang understand Qt semantics. It displays Qt related compiler warnings, ranging from unnecessary memory allocation to the misuse of APIs, and provides refactoring actions for fixing some of the issues. Clang tools are delivered and installed with Qt Creator, and therefore you do not need to set them up separately.

**CLAZY\_CHECKS** is a comma-separated list of checkers or check-sets to run. By default, all checkers from the 'level0' and 'level1' check-sets will run. Here are a few examples.

- To enable the two checkers 'unneeded-cast' and 'virtual-call-ctor' only, type:

```
% export CLAZY_CHECKS="unneeded-cast,virtual-call-ctor"
```

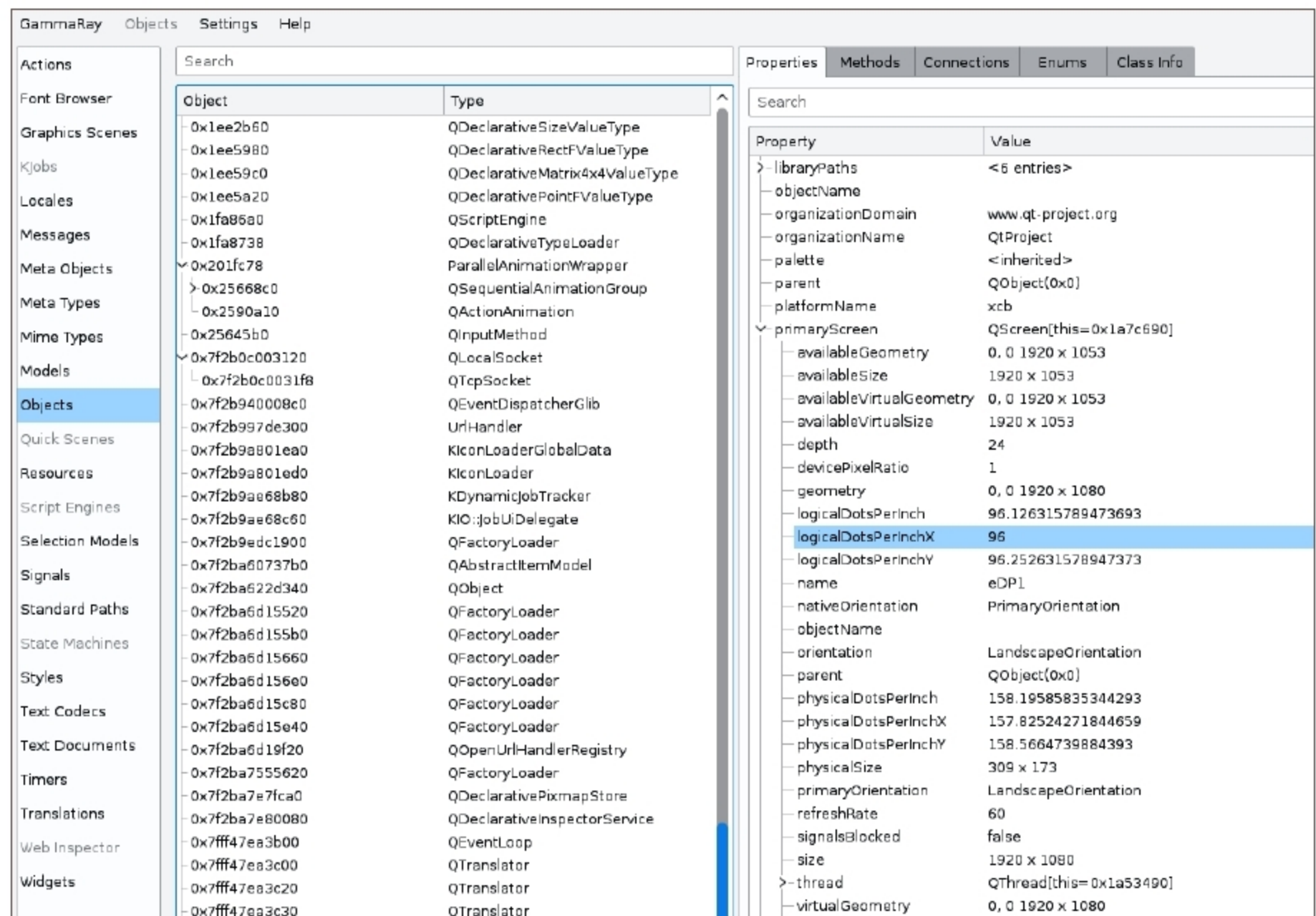


Figure 1: The GammaRay interface